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REPORT

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SECTION 1: COMPANY INTRODUCTION

1.1 Introduction to the Business

Caliber is a bespoke furniture design and manufacturing company based in the 6th of October City, Cairo, Egypt. The company specialises in the design and production of high-end, custom-made furniture and interior pieces, serving both individual clients and business-to-business (B2B) partners. Caliber occupies a niche within the Egyptian furniture market by offering a fully integrated service model: from initial concept and 3D design through to in-house fabrication, finishing, and delivery.

The company operates across two primary environments — a design office where product concepts are developed and technical drawings are produced, and a factory/workshop where the physical manufacturing takes place. This dual-environment structure allows Caliber to maintain tight quality control over its output while offering a high degree of customisation to clients.

I was placed at Caliber as a Product Designer intern, working across both the design and production sides of the business. My role involved end-to-end ownership of projects: from 3D modelling and technical drawing production to overseeing the manufacturing of the finished pieces on the factory floor.

1.2 The Organisation

Caliber was established as a small-to-medium enterprise (SME) operating in the premium custom furniture segment. Its client base includes interior designers, renovation studios, hospitality venues, and private residential clients. The company's competitive advantage lies in its ability to translate unique client briefs into fully custom, production-ready furniture pieces using a combination of advanced digital design tools and skilled craftsmanship.

The organisational structure of Caliber is relatively flat. Mr. Saad Kamar, who served as my Industrial Supervisor, oversees both the design and production operations. Under his leadership, a small team of designers, factory workers, blacksmiths, and finishing specialists collaborate to execute each project. This lean structure meant that as an intern, I was given direct access to decision-makers and was expected to operate with a significant degree of independence from an early stage.

The company's products span a wide range of furniture typologies including chairs, tables, benches, shelving, lighting, and custom installation pieces. Materials commonly used include stainless steel, sheet metal, iron, chrome-plated tube, solid wood, MDF, glass, resin, and a variety of upholstery fabrics and leathers.

1.3 The Environment

Caliber operates within the Egyptian luxury and semi-luxury furniture market, a sector that has experienced growing demand in recent years driven by an expanding real estate sector and a rising number of hospitality and commercial interior projects. The 6th of October City, where Caliber is located, is a major industrial and commercial hub west of Cairo, making it strategically positioned for manufacturing operations.

The competitive landscape in this market includes both local manufacturers and international brands. Caliber differentiates itself through its B2B model, customisation capability, and the integration of digital design tools (CAD/3D modelling) into its production workflow — a feature that is not universally adopted by competitors in the Egyptian market.

Technology plays a central role in Caliber's operations. The adoption of software such as Fusion 360, Shapr3D, and SolidWorks, combined with laser cutting technology and CNC-adjacent processes, positions Caliber at the more technically advanced end of the local furniture manufacturing spectrum. One challenge identified during the internship was the absence of a fully systematised manufacturing process, which occasionally led to inefficiencies on the production floor. I proactively proposed process improvements to help streamline production cycles.

1.4 The Functions

Caliber's operations are organised around two core functional areas:

Design & Engineering: This function is responsible for conceptualising products, producing 3D models, generating technical drawings, preparing DXF files for laser cutting, and managing client briefs. As a Product Designer intern, I was embedded within this function.

Manufacturing & Production: This function encompasses all physical fabrication activities including laser cutting, tube bending, welding, blacksmithing, powder coating, painting, upholstery, and final assembly. The factory operates on a project-by-project basis, with production timelines driven by client order deadlines.

A third, more informal function is sales and client management, which involves taking client briefs, producing quotations, and managing delivery. I was exposed to aspects of this function through my involvement in B2B order management, material sourcing, and client-driven customisation projects.

1.5 The Department / Section

I was attached to the Design Department / Product Design section of Caliber. Within this department, my responsibilities covered the full design-to-production pipeline for

assigned projects. On any given project, my workflow would typically begin with receiving a client brief or supervisor instruction, followed by conceptual development, 3D modelling, technical drawing production, DXF file preparation, and handoff to the manufacturing team — after which I would move onto the factory floor to supervise and quality-check the execution.

This holistic involvement across the entire product lifecycle — from digital design to physical realisation — was a defining feature of my internship and provided me with a depth of understanding that purely design-focused or purely production-focused roles would not have offered.

1.6 Communications and Information Technology

Communication within Caliber was largely informal and direct, reflecting the company's flat organisational structure. Day-to-day project communication between the design and production teams was handled in person, with supplementary communication via WhatsApp messaging — including the sharing of technical drawings and DXF files with factory supervisors and workshop partners.

On the technology side, the design workflow was built around industry-standard CAD and 3D modelling software: Fusion 360, Shapr3D, and SolidWorks. These tools were used to develop 3D models, extract technical drawings with full dimensional annotation, and generate DXF files for laser cutting. The use of digital files significantly improved the accuracy and speed of manufacturing compared to hand-drawn or loosely specified briefs.

One area where I observed a gap was in the absence of a formal project management or workflow tracking system. Communication was effective on a team level but lacked a structured record-keeping mechanism, which I identified as an area for potential process improvement.

1.7 Financial Structure

As a privately held SME, Caliber's detailed financial information was not disclosed during the internship. However, from my involvement in B2B order management and project quotation activities, I was able to observe several aspects of the company's financial operations.

Revenue is generated primarily through client orders, with pricing based on material costs, production complexity, and delivery timelines. B2B clients typically place bulk orders (e.g., 9 coffee tables, 16 chairs, 14 dining chairs), which represent significant revenue events for the company. I was involved in preparing and pricing customised products in coordination with the factory and workshop, including material selection, production methods, and cost breakdowns to support accurate project quotations.

The company's cost structure is driven by raw material procurement (stainless steel, sheet metal, chrome tube, fabrics, glass), labour costs across the design and production teams, and outsourced specialist services (e.g., resin mould workshops and external fabricators for specific processes).

1.8 Introduction to the Nature of Work

My internship at Caliber spanned 16 weeks from 15 January 2026 to 15 May 2026. The nature of work was project-driven, with each week typically centred on one or more active client orders. I was responsible for managing assigned projects from initial design concept through to final quality-checked delivery.

Key work activities throughout the internship included:

- 3D modelling of furniture pieces using Fusion 360, Shapr3D, and SolidWorks
- Production of detailed technical drawings with full dimensional annotation
- Generation of DXF files for laser cutting sheet metal and tube components
- B2B order management: translating client requirements into manufacturable designs
- Inventory checking and material sourcing
- Factory floor supervision during manufacturing
- Quality control checks prior to product delivery
- Material and finish specification (metals, fabrics, leathers, coatings)
- Coordination with external workshops for specialist processes
- Freelance furniture collection design for an external client studio
- Process improvement recommendations for the manufacturing workflow

1.9 Tasks – Application of Concepts and Knowledge

Throughout the internship, I was able to directly apply theoretical knowledge gained during my academic studies in Industrial Design to real-world professional challenges. The following highlights the most significant projects undertaken and the academic concepts applied:

Weeks 1–2: Orientation and Initial Design Work

During the first two weeks, I familiarised myself with the company's design and production workflow. I began producing 3D models and technical drawings using Fusion 360, Shapr3D, and SolidWorks — applying CAD skills developed in university coursework. I also prepared DXF files for laser cutting, directly applying knowledge of manufacturing file formats and tolerances. My first B2B order involvement required me to translate loosely defined client requirements into precise, manufacturable designs, applying design communication and technical drawing principles.

Weeks 3–4: B2B Order Management and Re-engineering

In Week 3, I managed a full B2B order from brief to production handoff, conducting inventory checks and coordinating with the production team to ensure on-time delivery. In Week 4, I re-engineered existing furniture items into new 3D models with updated technical drawings — applying reverse-engineering principles and critical design thinking to improve upon existing products.

Week 5: Stainless Steel Coffee Table Production (9 Units)

I designed and oversaw the production of 9 stainless steel coffee tables for a B2B client. This project applied my knowledge of sheet metal design, structural integrity, laser cutting processes, and quality assurance. The pre-polishing stage and final QC check required an understanding of surface finish standards and manufacturing tolerances.

Week 6: Lounge Chair and Buffet – Complex Manufacturing

Working on a lounge chair that required precise bending and cutting, I visited the factory to oversee the process and ensure accuracy. This week introduced me to additional machinery on the factory floor, expanding my practical manufacturing knowledge. The project applied ergonomics knowledge (chair dimensions, proportions) alongside manufacturing supervision.

Week 7: Material Sourcing and Innovation

My supervisor tasked me with sourcing materials and manufacturers for a glass-top table concept. Recognising that solid glass would be impractically heavy, I proposed using resin as an alternative material — applying problem-solving and materials science knowledge. I successfully identified an external workshop capable of producing large resin moulds, demonstrating initiative and supply chain awareness.

Week 8: Wooden Chair B2B Order (11 Units)

I re-engineered a coffee table design and oversaw a B2B order of 11 laser-cut wooden chairs assembled using a layered lamination technique. This project applied my understanding of wood manufacturing processes, laser cutting design for CNC production, and quality control for batch orders.

Week 9: Fabric Application and Surface Finishing (16 Chairs)

Working on the lounge chair designed in Week 4, I selected appropriate leather, oversaw the welding gap-filling and powder-coat painting process, and developed a wooden backrest substrate to allow fabric upholstery on a metal frame. This project significantly expanded my knowledge of multi-material finishing processes and upholstery techniques.

Part 2 – Week 1: Round Glass-Top Coffee Tables (5 Units)

I 3D-modelled a chrome tubular base for 5 round coffee tables in Shapr3D, engineering the arm geometry to ensure structural integrity. I supervised tube bending, welding, polishing, and glass mounting, performing final QC on all units before delivery.

Part 2 – Week 2: Two Chair Designs (7 + 14 Units)

I modelled and coordinated production for two chair designs simultaneously — a compact lounge chair and a dining/accent chair — demonstrating the ability to manage multiple concurrent projects and apply different structural and upholstery design principles to each piece.

Part 2 – Week 3: Technical Drawing Production

I produced a complete set of manufacturing-standard technical drawings for an ongoing furniture order, covering multiple components including panels, structural elements, and seating base frames with full dimensional annotation.

Part 2 – Week 4: Book Stand Design and Prototype

I designed a geometric sheet metal book stand from concept through to a first physical prototype. The prototype was fabricated in raw steel to validate the geometry and proportions before full production — applying prototype testing methodology and iterative design principles.

Part 2 – Week 5: Ceiling-Suspended DJ Table

One of the most technically demanding projects of the internship: a stainless steel DJ table suspended from a ceiling via steel rods and chains. I modelled the platform in SolidWorks, integrating the suspension system into the structural design, coordinated fabrication, and supervised on-site installation. This project applied structural engineering principles alongside interior design considerations.

Part 2 – Weeks 6–7: Freelance Furniture Collection Design

I designed a complete cohesive furniture collection for an external creative studio — including a lounge chair, bench, side table, coffee table, cylindrical stool, and floor lamp — sharing a teal/chrome design language. In the second phase, I refined the collection by introducing an arch motif inspired by Islamic and Moroccan architectural forms, producing final high-quality renders for client presentation.

Part 2 – Week 8: Custom Curved Iron Chair (4 Units)

I designed a classically inspired armchair with a sculptural scrolled X-frame base in Shapr3D, modelling complex tube bending paths to ensure geometrical consistency and manufacturability. Four units were produced under my supervision.

Part 2 – Week 9: 3D Modelling for Upcoming Production

In the final week, I modelled a mid-century lounge chair with an angular wooden X-frame base and a dining table with a tapered pyramid base — both prepared to technical drawing extraction standard and handed off to production as a deliverable for future orders.

1.10 Development of Employability Skills

The 16-week internship at Caliber was a transformative experience that significantly accelerated my development across a wide range of employability skills. The following outlines the key areas of growth:

Technical Design Skills

My proficiency in 3D modelling software — particularly Shapr3D, Fusion 360, and SolidWorks — was elevated substantially through daily professional application. The speed and accuracy with which I could produce models and technical drawings improved week-on-week, and I developed a much more nuanced understanding of design-for-manufacture constraints. My supervisor rated my technical knowledge as 'A' (Excellent) in his end-of-internship assessment.

Manufacturing Knowledge

Working on the factory floor exposed me to manufacturing processes that are rarely taught practically in university: tube bending, welding, powder coating, laser cutting file preparation, layered wood lamination, resin mould sourcing, and fabric upholstery on metal frames. This hands-on exposure represents a significant addition to my skill set as a product designer.

Project Management and Order Coordination

Managing B2B orders end-to-end — from brief interpretation through design, inventory checking, production coordination, and quality control — developed my project management capabilities considerably. I learned how to balance design quality with time and resource constraints in a real commercial environment.

Problem-Solving and Initiative

Throughout the internship I was consistently placed in situations that required creative problem-solving: proposing resin as an alternative to glass, developing wooden substrates for fabric application on metal, suggesting process improvements to the manufacturing workflow, and sourcing specialist external workshops. My supervisor noted: 'Very very good for his age, always searches for new solutions.'

Communication and Leadership

Working directly with factory workers, blacksmiths, and external workshop partners required clear, effective communication of technical requirements. I learned to convey complex design intentions to non-design professionals in practical, actionable terms. My supervisor noted my strong oral communication skills (rated 'A') while also identifying that I tend to take a leadership role — a strength that I am learning to balance with collaborative flexibility.

Professional Conduct

Operating within a professional environment with real client deliverables and financial consequences developed my understanding of workplace standards, timekeeping, confidentiality, and professional accountability. The experience of receiving frank supervisor feedback — including the observation that I can be less receptive to direct instructions — has provided me with important self-awareness for my professional development.

SECTION 2: FUTURE PLANS FOR EMPLOYMENT

2.1 Updated Plan for Employment and Continuing Professional Development

The internship at Caliber has significantly refined and focused my career aspirations. Prior to the placement, my career goals were broadly oriented towards product design; the 16 weeks at Caliber have crystallised this into a specific ambition to work at the intersection of furniture design, metal fabrication, and high-end interior product development.

In the short term (0–2 years), my goal is to complete my degree at APU and secure a junior product designer position within a design studio or furniture manufacturer that operates at the premium/custom segment of the market. The experience gained at Caliber — particularly in B2B project management, 3D modelling for manufacture, and factory supervision — has given me a strong and differentiated portfolio that I will continue to build and refine. I intend to develop my skills in product visualisation (rendering and photography) to better communicate design proposals to clients, and to deepen my proficiency in SolidWorks, which I identified as the most powerful tool for complex structural design during the internship.

In the medium term (2–5 years), I aim to progress towards a senior designer or design lead role, taking greater responsibility for creative direction on projects. Given my interest in materials and manufacturing, I would also like to pursue continuing professional development in advanced manufacturing techniques — particularly CNC machining, metal casting, and advanced composite materials — to expand the range of design solutions I can offer. Courses and certifications from industry bodies such as the Chartered Society of Designers (CSD) would support this development.

In the longer term, my ambition is to establish my own design practice or furniture brand, drawing on the end-to-end production knowledge gained at Caliber. The ability to design, engineer, and manufacture — rather than just design — is a competitive advantage that I am committed to developing throughout my career.

2.2 Plan for Securing Good Quality Internships for Future Students

Based on my internship experience at Caliber, the following recommendations are offered to assist future students in securing high-quality industrial placements in product design and related fields:

Build a Practical Portfolio Before Applying

Many design companies — particularly in the manufacturing sector — prioritise demonstrable practical skills over academic credentials alone. Future students should develop a portfolio that includes 3D models, technical drawings, and rendered concepts

produced using industry-standard tools (Fusion 360, SolidWorks, Shapr3D, Rhino). Employers like Caliber respond strongly to applicants who can show they can 'hit the ground running.'

Target Manufacturing-Integrated Design Companies

For product design students, placements at companies that combine design with in-house manufacturing provide far richer learning experiences than purely studio-based environments. Seek out furniture manufacturers, metal fabricators, and custom interior product companies that have both design and production capabilities.

Communicate Proactively with the Industrial Supervisor

From the first week, establish clear communication with the Industrial Supervisor about expectations, workload, and the types of projects you would like to be involved in. Supervisors appreciate interns who show initiative and actively seek out opportunities rather than waiting to be assigned work.

Embrace the Factory Floor

One of the most valuable aspects of my internship was time spent on the factory floor. Future students should not limit themselves to the design office — understanding how designs are physically realised is essential for becoming a well-rounded product designer.

Document Everything Consistently

Maintaining a thorough and up-to-date logbook from the first week makes the final report submission significantly easier. Photograph your work regularly, note down key decisions and lessons learned, and keep copies of all technical drawings produced.

2.3 Conclusions and Recommendations

The 16-week internship at Caliber has been one of the most formative experiences of my academic career. Working as a Product Designer in a real commercial environment — managing client orders, producing professional technical documentation, supervising manufacturing, and delivering finished products — has provided a depth and breadth of practical education that complements and in many ways surpasses what can be achieved in a university setting alone.

The key conclusions drawn from this experience are as follows:

- Design and manufacturing are inseparable disciplines. A product designer who understands manufacturing processes at a practical level produces better, more manufacturable designs.
- Soft skills are as important as technical skills. The ability to communicate complex design information to non-designers, to manage time under commercial pressure, and to accept and act on critical feedback are all essential professional competencies.

- Initiative and problem-solving are highly valued in professional settings. The projects where I added the most value were those where I identified a problem and proposed a solution proactively — such as the resin material substitution and the process improvement recommendations.
- Self-awareness is a critical component of professional development. The feedback from my Industrial Supervisor — particularly regarding attitude to supervision and ego management — has given me valuable self-awareness that I will carry forward into my professional career.

Recommendations for APU:

- Introduce more hands-on manufacturing workshops, particularly in metalworking (sheet metal, welding, tube bending). The current university workshop provision is predominantly wood-focused, which limits the preparation of product design students for industries that work with metal and glass.
- Integrate design-for-manufacture principles more deeply into the curriculum, with real constraints (material cost, production time, batch quantities) included in project briefs.
- Encourage students to develop proficiency in multiple CAD platforms (particularly SolidWorks and Fusion 360) before entering the workplace, as these are the most commonly required tools in product and furniture design companies.

Overall, I would strongly recommend Caliber as a placement company for future APU product design students. The breadth of project exposure, the quality of practical learning, and the professional culture at Caliber make it an exceptional environment for an industrial design intern to develop both technical and professional skills.

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